MATH317 · NUMERICAL METHODS

Mock Midterm

اختبار تجريبي بنفس شكل الميد · 60 دقيقة · 30 درجة

محارب

01 الأسئلة

TRUE / FALSE · 7 M — Q1

1) The relative error is the absolute error divided by the approximate value.
2) Chopping to n digits discards all digits after the n-th digit.
3) A Maclaurin series is a Taylor series expanded about c = 1.
4) In single precision the bias of the exponent is 127.
5) Machine epsilon is the largest number that a computer can store.
6) In the bisection method the error bound is halved at every step.
7) Newton's method needs both f(x) and f′(x) at every step.

HORNER + TAYLOR · 3 + 5 M

Q2

Given $P(x) = 2x^3 - 5x^2 + 7$.

a) Use Horner's algorithm to evaluate $P(2)$.

b) Using the complete Horner's algorithm, find the Taylor expansion of $P(x)$ about $r = 1$.

FLOATING-POINT · 7 M

Q3

Determine the single-precision machine representation in a 32-bit word-length computer of the decimal number −37.40625.

BISECTION · 2 + 6 M

Q4

Given $f(x) = x^3 + x - 3$.

a) Prove that f has exactly one real root in the interval [1, 2].

b) Find a root on [1, 2] using the bisection method. Terminate the iterations if they exceed nmax = 2 or if the error bound is less than 0.5×10^{-6}.

NEWTON · BONUS

Q5

For the same $f(x) = x^3 + x - 3$, start from $x_0 = 1$ and apply two iterations of Newton's method.